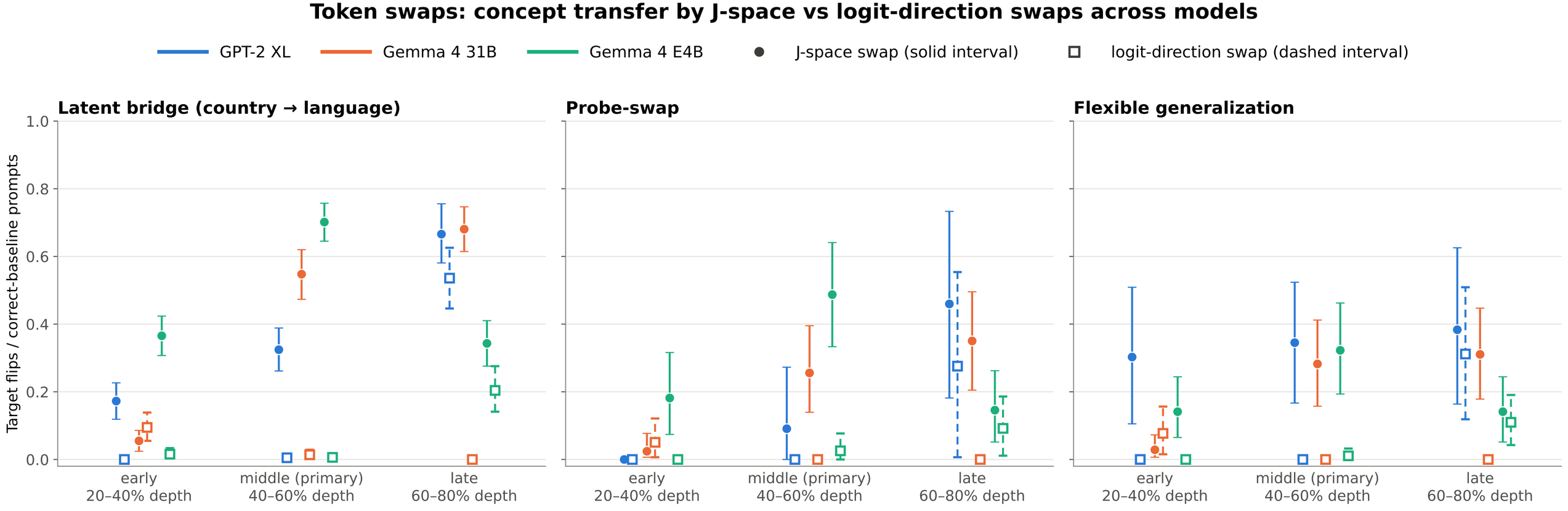
Token swaps: concept transfer by J-space vs logit-direction swaps across models
GPT-2 XL
Gemma 4 31B
Gemma 4 E4B
J-space swap (solid interval)
logit-direction swap (dashed interval)
Latent bridge (country → language)
Probe-swap
Flexible generalization
Target flips / correct-baseline prompts
1.0
0.8
0.6
0.4
0.2
0.0
early
20–40% depth
middle (primary)
40–60% depth
late
60–80% depth
α = 1; markers = success rate among prompts with a correct baseline, vertical bars = pointwise 95% prompt-clustered bootstrap intervals (10,000 resamples; none where degenerate).
Norm-matched random swaps stay at or below 2% in every band (not shown). Primary band exact; other bands read from the notebooks' charts (error at most 0.01).